

Re: Final Project Presentation – Policy Topic Proposal

From: Zachary Johnigan zacharyjohnigan@uchicago.edu

To: Aida Pacheco-Applegate apacheco@uchicago.edu

Date: Thu, Aug 13, 2026, 11:33 AM



Final_Project_Presentation.Rmd 7 KB

mass_shootings.csv 43 KB

Dear Professor,

I wanted to share my **proposed topic and initial approach** for the Final Project Presentation.

Proposed Policy Area: Mass Shootings in the United States

I plan to focus on **mass shootings in the United States**, using the mass_shootings dataset that I previously worked with in Computing for the Social Sciences. The dataset contains **125 observations and 14 variables** describing mass-shooting incidents.

Initial Research Approach

My initial research approach is to examine two variables in particular:

- location — where mass shootings occur geographically
- location_type — the types of environments in which they occur, such as workplaces, schools, or other locations

I would like to explore whether there are observable patterns in either of these variables.

Moving from the Data to a Specific Policy Issue; If the exploratory analysis identifies a notable pattern, I would then narrow the project to a specific policy issue associated with that pattern.

For example, if the data showed a **concentration of incidents in a particular state and/or type of environment**, I would investigate gun-safety policies that have been proposed, implemented, or debated in that context. This would allow the quantitative analysis to help guide the selection of a more specific policy question rather than selecting one in advance.

If location and location_type do not reveal a meaningful pattern, I plan to explore other variables in the dataset to identify a more appropriate relationship for the final analysis.

R Code, Dataset, and Repository

I have also begun preparing and cleaning the dataset in R using techniques from our labs. I am documenting the work in my Final_Project_Presentation.Rmd file and plan to maintain the project in my course GitHub repository so that the code and development of the analysis are available for review.

Here is a direct link to the project folder if you would like to review the dataset and code:

Final Project Presentation — GitHub Repository

https://github.com/ZacharyJohnniganAcademia/2_Data_For_Policy_Analysis_-_Mgmt/tree/main/Final_Project_Presentation

If you are unable to view the files through GitHub, I have also attached them.

Before I proceed further, I wanted to confirm that this exploratory approach and the broader policy area of mass shootings in the United States would be appropriate for the Final Project Presentation.

Best,
Zachary Johnnigan

From: Aida Pacheco-Applegate apacheco@uchicago.edu

To: Zachary Johnnigan zacharyjohnnigan@uchicago.edu

Date: Thu, Aug 13, 2026, 6:37 PM

Hi Zachary,

Thanks for sending this along! I think this is an interesting topic, and I like your idea of starting with the data and then narrowing down the policy question.

One thing I would like you to think about is the structure of your data. Since the dataset includes incidents from the 1990s through 2021, you have multiple observations over time and potentially multiple observations for the same location. Because we have not covered longitudinal/panel data analysis in class, I would not recommend trying to analyze the changes over time directly.

Instead, you could potentially aggregate the incident-level data. For example, if you are interested in geographic differences, you could aggregate the number of incidents by state across the entire study period, giving you one observation per state. You could then think about whether there is another variable you could use to explore a relationship across states.

I would also encourage you to think beyond simply describing `location` and `location\_type`. What relationship between variables are you interested in understanding? For example, could you use information about the characteristics of the incidents to ask a question that can be answered using the methods we have covered in class?

Your general approach of exploring the data first makes sense to me. I just want to make sure that, given the way the data are structured, you end up with a question and analysis that you can answer using the methods we have learned in class.

Take a look at the data with this in mind, and let me know if you have more questions.

Best,
Aida

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